

NEGATIVITY LEMMA

1. DETAILED ARGUMENT

Lemma 1.1. *Let $f: X \rightarrow Y$ be a projective birational morphism between normal varieties, with Y affine. Then there exists an effective Cartier divisor E on X such that $-E$ is f -ample.*

Proof. Since f is projective, we can choose an f -ample Cartier divisor D on X . Put $\mathcal{F} = f_*\mathcal{O}_X(-D)$. Since f is projective, $\mathcal{F}$ is coherent. Since f is birational, it is an isomorphism over a dense open subset of Y , on which $\mathcal{F}$ is invertible. Thus $\mathcal{F}$ has generic rank one; in particular, $\mathcal{F} \neq 0$.

Since Y is affine, a nonzero quasi-coherent sheaf on Y has a nonzero global section. Hence we may choose

$$0 \neq s \in H^0(Y, \mathcal{F}) = H^0(X, \mathcal{O}_X(-D)).$$

Viewing $\mathcal{O}_X(-D)$ as a subsheaf of the sheaf of rational functions, we regard s as a nonzero rational function on X . Set

$$E = -D + \operatorname{div}(s).$$

The condition $s \in H^0(X, \mathcal{O}_X(-D))$ says precisely that E is an effective Cartier divisor. Indeed, if d is a local equation of D , then s/d is a nonzero regular function and is a local equation of E . Finally, $-E \sim D$, so $-E$ is f -ample. $\square$

Lemma 1.2. *Let $f: X \rightarrow Y$ be a projective birational morphism between varieties, with Y normal, and let E be an effective Cartier divisor on X such that $-E$ is f -ample. Then $\operatorname{Supp} E$ contains the exceptional locus $\operatorname{Exc}(f)$.*

Proof. Since Y is normal, $\operatorname{Exc}(f)$ is covered by complete integral curves contracted by f . For every such curve C , the f -ampleness of $-E$ gives $E \cdot C < 0$. Since E is effective, this forces $C \subseteq \operatorname{Supp} E$. Hence $\operatorname{Exc}(f) \subseteq \operatorname{Supp} E$. $\square$

Theorem 1.3 (Projective negativity lemma). *Let $f: X \rightarrow Y$ be a projective birational morphism between normal varieties. Let D be an $\mathbb{R}$ -Cartier divisor on X such that $-D$ is nef over Y . Then:*

- (1) D is effective if and only if f_*D is effective.
- (2) If D is effective, then for every $y \in Y$, either

$$f^{-1}(y) \cap \operatorname{Supp} D = \emptyset \quad \text{or} \quad f^{-1}(y) \subset \operatorname{Supp} D.$$

Proof of Theorem 1.3. Both assertions are local on Y , so we may assume that Y is affine.

(1) One direction is trivial. For the other direction, assume $f_*D \geq 0$. Since f is birational, every prime divisor with negative coefficient in D is f -exceptional.

Suppose, for a contradiction, that D is not effective. Choose an effective Cartier divisor E on X such that $-E$ is f -ample, as in Lemma 1.1. By Lemma 1.2, the coefficient $\text{coeff}_P(E) > 0$ whenever $\text{coeff}_P(D) < 0$. Set

$$e := \max_{\text{coeff}_P(D) < 0} \frac{-\text{coeff}_P(D)}{\text{coeff}_P(E)} > 0.$$

Then $D + eE \geq 0$, and we can find an f -exceptional prime divisor F such that $F \not\subset \text{Supp}(D + eE)$. Choose a closed point $x \in F \setminus \text{Supp}(D + eE)$ and a curve $C \subset F$ through x contracted by f . Then $C \not\subset \text{Supp}(D + eE)$, so $(D + eE) \cdot C \geq 0$. On the other hand,

$$(D + eE) \cdot C = D \cdot C + e(E \cdot C) < 0,$$

because $-D$ is f -nef, $-E$ is f -ample, and $e > 0$. This is a contradiction, so $D \geq 0$.

(2) Assume $D \geq 0$. Since f is projective and birational and Y is normal, its fibers are connected by Stein factorization. Suppose $f^{-1}(y) \cap \text{Supp } D \neq \emptyset$. If $f^{-1}(y) \not\subset \text{Supp } D$, connectedness gives a curve $C \subset f^{-1}(y)$ meeting $\text{Supp } D$ but not contained in it. Then $D \cdot C > 0$, contradicting the f -nefness of $-D$. Thus $f^{-1}(y) \subset \text{Supp } D$. $\square$

Theorem 1.4 (Proper negativity lemma). *Let $f: X \rightarrow Y$ be a proper birational morphism between normal varieties. Let D be an $\mathbb{R}$ -Cartier divisor on X such that $-D$ is nef over Y . Then:*

- (1) D is effective if and only if f_*D is effective.
- (2) If D is effective, then for every $y \in Y$, either

$$f^{-1}(y) \cap \text{Supp } D = \emptyset \quad \text{or} \quad f^{-1}(y) \subset \text{Supp } D.$$

Proof. By Chow's lemma and normalization, choose a proper birational morphism $\pi: X' \rightarrow X$ with X' normal such that $f' = f \circ \pi: X' \rightarrow Y$ is projective. Set $D' = \pi^*D$. By the projection formula, $-D'$ is f' -nef, and

$$\pi_*D' = D, \quad f'_*D' = f_*D.$$

If $f_*D \geq 0$, Theorem 1.3(1) gives $D' \geq 0$, hence $D = \pi_*D' \geq 0$. The converse is immediate.

If $D \geq 0$, then $D' \geq 0$ and $\text{Supp } D' = \pi^{-1}(\text{Supp } D)$. By Theorem 1.3(2), each fiber of f' is either disjoint from $\text{Supp } D'$ or contained in it. Since $f'^{-1}(y) \rightarrow f^{-1}(y)$ is surjective, the same holds for f and D . $\square$

2. A CONCISE PROOF

Proof of Theorem 1.4. We may work over an affine open subset of Y . By Chow's lemma and normalization, choose $\pi: X' \rightarrow X$ proper birational with X' normal and $f \circ \pi$ projective. The projection formula preserves relative nefness, and

$$\pi_*\pi^*D = D, \quad \text{Supp}(\pi^*D) = \pi^{-1}(\text{Supp } D) \quad \text{if } D \geq 0.$$

Thus both assertions descend from X' to X , and we may assume that f is projective.

(1) Suppose $f_*D \geq 0$ but $D \not\geq 0$. Choose an f -ample Cartier divisor A . The coherent sheaf $f_*\mathcal{O}_X(-A)$ has generic rank one and hence, since Y is affine, a nonzero global section s . Then $E = -A + \text{div}(s) \geq 0$ and $-E$ is f -ample. Every contracted curve C satisfies $E \cdot C < 0$, so $C \subset \text{Supp } E$. Such curves cover $\text{Exc}(f)$; hence $\text{Exc}(f) \subset \text{Supp } E$.

All negative components of D are f -exceptional. Set

$$e := \max_{\text{coeff}_P(D) < 0} \frac{-\text{coeff}_P(D)}{\text{coeff}_P(E)} > 0.$$

Then $D + eE \geq 0$, and some f -exceptional prime divisor F is not contained in $\text{Supp}(D + eE)$. Choose a contracted curve $C \subset F$ not contained in this support. We obtain

$$0 \leq (D + eE) \cdot C = D \cdot C + e(E \cdot C) < 0,$$

a contradiction. The reverse implication is immediate.

(2) Assume $D \geq 0$ and $f^{-1}(y) \cap \text{Supp } D \neq \emptyset$. The fiber is connected by Stein factorization. If it is not contained in $\text{Supp } D$, there is a curve in the fiber meeting $\text{Supp } D$ but not contained in it. Its intersection with D is positive, contrary to the f -nefness of $-D$. Thus $f^{-1}(y) \subset \text{Supp } D$. $\square$

3. EXTENSION TO QUASI-PROJECTIVE BASES

Lemma 1.1 remains valid when the affine hypothesis on Y is replaced by quasi-projectivity: for a projective birational morphism $f: X \rightarrow Y$ between normal varieties with Y quasi-projective, there is an effective Cartier divisor E such that $-E$ is f -ample. This strengthening also requires no Riemann–Roch.

Proof. Choose an f -ample line bundle $\mathcal{L}$ and put $\mathcal{F} = f_*\mathcal{L}^{-1}$. As in Lemma 1.1, $\mathcal{F}$ is coherent of generic rank one, so $\mathcal{F} \neq 0$.

Choose a very ample Cartier divisor H on Y . By the global-generation criterion for ampleness, $\mathcal{F} \otimes \mathcal{O}_Y(mH)$ is generated by global sections for $m \gg 0$. It is nonzero, so it has a nonzero global section. The projection formula gives

$$\begin{aligned} 0 \neq s &\in H^0(Y, \mathcal{F} \otimes \mathcal{O}_Y(mH)) \\ &= H^0(X, \mathcal{L}^{-1} \otimes \mathcal{O}_X(mf^*H)). \end{aligned}$$

Its zero divisor E satisfies

$$\mathcal{O}_X(-E) \simeq \mathcal{L} \otimes f^* \mathcal{O}_Y(-mH).$$

Tensoring with a line bundle pulled back from the base preserves relative ampleness, so $-E$ is again f -ample. If Y is projective, then X is projective. Taking $\mathcal{L} = \mathcal{O}_X(A)$ for any very ample divisor A on X gives $E \sim mf^*H - A$ and hence the decomposition

$$mf^*H \sim A + E, \quad -E \sim_f A.$$

Lemma 1.2 further shows that every exceptional prime divisor lies in $\text{Supp } E$. The only additional input is global generation after a sufficiently large ample twist. No Riemann–Roch or section-growth estimate is needed. $\square$

Remark 3.1. The affine hypothesis is essential to the *untwisted argument*, not to the conclusion over a quasi-projective base. A nonzero coherent sheaf on a projective variety need not have any global sections. For example, with $f = \text{id}_{\mathbf{P}^1}$ and $\mathcal{L} = \mathcal{O}_{\mathbf{P}^1}(1)$, one has $f_*\mathcal{L}^{-1} = \mathcal{O}_{\mathbf{P}^1}(-1) \neq 0$ but $H^0(X, \mathcal{L}^{-1}) = 0$. The twist by mH repairs precisely this failure. On an arbitrary normal base, a global very ample H need not exist; for the negativity lemma it suffices to apply the affine construction on each affine open subset of the base.

Remark 3.2. For comparison, the projective decomposition also has the following proof, making explicit the section-growth argument usually used in Kodaira’s lemma. Assume that Y is projective, put $n = \dim X = \dim Y \geq 1$, choose very ample Cartier divisors H on Y and A on X , and set $M = f^*H$. Since Y is normal and f is proper birational, $f_*\mathcal{O}_X = \mathcal{O}_Y$. Thus

$$h^0(X, \mathcal{O}_X(mM)) = h^0(Y, \mathcal{O}_Y(mH)) = \frac{H^n}{n!} m^n + O(m^{n-1}), \quad H^n > 0.$$

The last equality follows from Serre vanishing and the Hilbert polynomial of the very ample divisor H . In particular, M is big; it need not be ample.

Choose an effective member $T \in |A|$. It is a Cartier divisor on the integral variety X , and $\dim T = n - 1$. The coherent sheaf $\mathcal{G} = f_*\mathcal{O}_T$ is supported on $f(T)$, whose dimension is at most $n - 1$. By the projection formula and the Hilbert polynomial of $\mathcal{G}$ on Y ,

$$\begin{aligned} h^0(T, \mathcal{O}_T(mM)) &= h^0(Y, \mathcal{G} \otimes \mathcal{O}_Y(mH)) \\ &= O(m^{n-1}). \end{aligned}$$

No normality or irreducibility of T is needed. Taking global sections of the exact sequence

$$0 \longrightarrow \mathcal{O}_X(mM - T) \longrightarrow \mathcal{O}_X(mM) \longrightarrow \mathcal{O}_T(mM) \longrightarrow 0$$

gives

$$\begin{aligned} h^0(X, \mathcal{O}_X(mM - A)) &\geq h^0(X, \mathcal{O}_X(mM)) - h^0(T, \mathcal{O}_T(mM)) \\ &\geq \frac{H^n}{n!} m^n - O(m^{n-1}) > 0 \end{aligned}$$

for $m \gg 0$. We only use left exactness here; surjectivity of the restriction map is unnecessary. A nonzero section therefore gives an effective Cartier divisor $E \sim mM - A$. Equivalently,

$$f^*H \sim_{\mathbb{Q}} \frac{1}{m}A + \frac{1}{m}E, \quad -E \sim_f A.$$

This is the requested decomposition into an ample divisor and an effective divisor. Lemma 1.2 shows that $\text{Supp } E$ contains every exceptional prime divisor. When $n = 0$, the assertion is immediate with $E = 0$.

The essential numerical input in this second proof is that the space of sections on X grows like m^n , whereas the space of restrictions to T grows at most like m^{n-1} . It is often expressed as an asymptotic Riemann–Roch argument. However, full Riemann–Roch is *not logically necessary*: the Hilbert polynomial and Serre vanishing on Y already give both estimates, and the first proof needs only global generation after an ample twist. Riemann–Roch alone computes an Euler characteristic, not h^0 ; a proof invoking it must also justify the relevant cohomological estimates. In particular, one cannot apply Serre vanishing on X merely because f^*H is big and nef. Neither construction uses the Hodge index theorem, and E may have nonexceptional components.